

IoT Wi-Fi TEMPERATURE-LOGGING SYSTEM

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Here is a temperature data-logging system based on PIC16F887 microcontroller (MCU), Wi-Fi and ThingSpeak application programming interface (API). ThingSpeak is an open source Internet of Things (IoT) application and API to store and retrieve data from things over the Internet. It enables you to collect, store, analyse, visualise and act on the data received from sensors or electronic circuits.

Block diagram of the temperature-logging system is shown in Fig. 1. Author's prototype is shown in Fig. 2. Temperature data transmitted from the circuit board can be accessed on a smartphone

or PC through Wi-Fi using ThingSpeak API.

Circuit and working

Circuit diagram of the IoT Wi-Fi temperature-logging system is shown in Fig. 3. It is built around PIC16F887 MCU (IC4), ESP8266

range of 2.0V-5.5V. Its inbuilt A/D converter has a 10-bit resolution and three internal timers. It supports enhanced USART module, in-circuit serial programming (ICSP), 3-wire SPI and I2C.

ESP8266 Wi-Fi module.

ESP8266 Wi-Fi module (Fig. 4) is a self-contained system on chip (SoC) with an integrated TCP/IP protocol stack that can give any MCU access to your Wi-Fi network. ESP8266 is capable of either hosting an application or offloading all Wi-Fi networking functions from another application processor.

Each ESP8266 module comes pre-programmed with an AT command set firmware. This means that you can simply hook this up to your Arduino device and get about as much Wi-Fi ability as a Wi-Fi shield can offer (and that is just out of the box). ESP8266 module is an extremely cost-effective board with a huge and an ever-growing community.

This module requires a 3.3V DC regulated supply. Power consumption of the ESP module is around 250mA when transmitting. You need proper power supply. Do not power it directly from a 5V supply source. It also needs a level shifting/conversion to protect the 3.3V device from 5V.

Logic-level shifter. ESP8266 module is not capable of converting

Test Points	
Test point	Details
TP0	0V (GND)
TP1	5V
TP2	3.3V
TP3	Low when S1 is pressed
TP4	Low when S2 is pressed

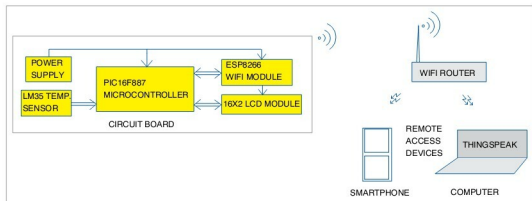


Fig. 1: Block diagram of the temperature-logging system

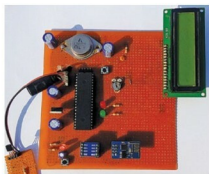


Fig. 2: Author's prototype

Wi-Fi module, logic-level shifter, LM35 temperature sensor (IC3), 3.3V regulator LM1117-3.3 (IC2), 5V regulator 7805 (IC1) and 16x2 LCD module (LCD1).

PIC16F887 MCU. This is a high-performance RISC CPU, 8-bit CMOS MCU, and has 8192-word flash memory, 368 bytes SRAM, 256 bytes EEPROM and 35 I/Os. It supports up to 20MHz oscillator/clock input and has a wide operating voltage